

# The Future: More Uses, More Models, More Scope

*Smart IDs are at a very crucial stage. The technology is neither nascent nor ubiquitous. It is in a mid stage where both technological developments and application-oriented innovations are happening in parallel—with each fuelling the other. Let us look at a few trends that hint at the future of smart cards.*

**Larger capacities.** Several governments are considering issuance of smart card-based IDs. Such cards would need large memories to store more details about the citizens. Plus, in the telecom world too, there is a convergence of device, content and services which requires the smart cards (SIMs) to be of a larger capacity. Hence the smart card players are vying with each other to launch larger-capacity cards.

**Increased application in automation and robotics.** Machine-to-machine (M2M) communications is one of the key applications fuelling the increase in demand of smart cards. M2M communications refers to communication between several machines or between machines and a server without any human intervention. It could be a network of sensors that measure various parameters such as temperature and humidity and communicate them to a server. Or, it could be a set of machines on a plant floor that communicate job or health status regularly to other machines for scheduling and self-diagnostic purposes. It could be smart meters communicating energy consumption information from a consumer's house to the power grid. Or, medical equipment relaying snapshots of a patient's health conditions to the hospital. Basically, smart cards make such communications much easier and more powerful. They also enable the use of existing devices and wireless networks for the same. A cellphone could act as a sensor, and the telecom network could be the carrier!

**Near-field communications (NFC).** This is a short-range, high-frequency communication technology that would enable your mobile phone to act as a smart card reader and writer, converting it into a very powerful device that could enable automatic billing, instantaneous payments, no-stop check-in at airports and more. The hype has been on for long. However, not much practical action has been seen at this front, perhaps because it needs significant investment in infrastructure, standards, ironing out of potential problems during deployment, etc. However, it is believed (according to recent research reports by various firms) that NFC will pick up momentum in the next few years and applications will go live. This, again, would give a boost to the smart card industry.

**A smart card in your watch.** The SIM card in your mobile phone is nothing but a smart ID card. Understandably, the mobile industry is one of the largest customers of the

smart card industry. The rise of 3G is expected to give smart card markets and technology a boom. Plus, the increase in the number of mobile devices ranging from mini, micro phones to handheld tablets and netbooks is also encouraging the industry to come up with more powerful, higher-capacity, high-featured yet small smart cards. The iPad has, in fact, led to sudden popularity of the 3FF or Micro SIM, which was actually introduced several years ago. A Micro SIM measures only 12×15 mm<sup>2</sup>. If it gets any smaller, you could have one in your watch!

**Authentication for remote banking.** Banks in developed countries have already replaced their magnetic swipe cards with smart cards. In other relatively less-developed countries, this operation is still pending—but another opportunity has also cropped up! Asian and African countries are showing a strong inclination towards mobile banking through modes such as SMS. Once NFC picks up, SMS+NFC could be a deadly combination for next-generation banking. Smart cards will play a key role in this revolution, as the need for foolproof authentication is very strong.

**Innovative uses in developing countries.** Developing countries like India can put smart cards to a lot of innovative uses once the basic infrastructure is in place. South Africa has some interesting examples in place. When a mining company near Port Elizabeth found that the petrol intended for their trucks was going amiss, they used smart cards near the mouth of the petrol tank to ensure that the petrol was filled into the designated vehicles alone and not stolen and sold to others. If the card is removed from a truck, it would alert the company and disable itself, so that it cannot be refitted on another truck.

When another town realised that parking attendants were getting mugged for the cash they carried on them, they decided to use smart cards as a parking solution instead. Another company used smart cards to monitor their rental car fleet, where customers could use rechargeable smart cards to access any of the company's rental cars parked close to their locality. They just have to find a car, swipe the card near a reader fitted on it, get in and drive away. ●

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